



Modelling Financial Markets

Chapter 1: Stylised Facts and Returns



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Acronyms used in this material (1/2)

- ACF: AutoCorrelation Function
- AI: Artificial Intelligence
- ARCH: AutoRegressive Conditional Heteroskedasticity
- ARCH-LM: ARCH Lagrange Multiplier (test)
- BET: Bucharest Exchange Trading (index)
- BET-TR: BET Total Return (index)
- BNR: Banca Națională a României (the National Bank of Romania)
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CAGR: Compound Annual Growth Rate
- CI: Confidence Interval
- CLT: Central Limit Theorem
- CRIX: CRypto IndeX
- DAX: Deutscher Aktienindex (the German stock index)
- EGARCH: Exponential GARCH
- ETF: Exchange-Traded Fund
- FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GEO: Government Emergency Ordinance
- GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- JB: Jarque–Bera (test)
- LB: Ljung–Box (test)
- MDD: Maximum DrawDown
- OHLC: Open, High, Low, Close
- PDV: Path-Dependent Volatility

Acronyms used in this material (2/2)

- ▣ **QQ: Quantile–Quantile (plot)**
- ▣ **RMSE: Root Mean Squared Error**
- ▣ **SE: Standard Error**
- ▣ **SSRN: Social Science Research Network**
- ▣ **TSPL: Time-Shifted Power Law (kernel)**
- ▣ **US: United States**
- ▣ **VIX: CBOE Volatility Index**

Outline

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What Is a Stylised Fact?

Definition

- ▣ A **stylised fact** is a statistical regularity observed across many assets, markets and periods
- ▣ It is qualitative, not quantitative: “returns have heavy tails”, not “kurtosis = 7.3”
- ▣ It describes **returns**, not price levels; the standard survey is [Cont \(2001\)](#)

Key message of this chapter

- ▣ Returns show little **linear autocorrelation**, but their **size** (absolute value, a proxy for risk) is highly structured

Why Stylised Facts Matter

- They are the **benchmark** every model of returns must reproduce
- A model that violates them misprices risk: margins, capital and option prices are wrong
- The textbook model of normally distributed, i.i.d. returns fails almost all of them
 - ▶ i.i.d. = **independent and identically distributed**: every day is a new, independent draw from the same distribution
- Stylised facts tell us **which** features a better model needs
- Later chapters build on them: volatility (5, 6, 9), risk (7, 8), machine learning and bubbles (13, 17)

Data Used in This Chapter

Series	Market	Period
S&P 500 index	US equities (price index)	1990–2026
BET-TR	Romanian equities (total return)	2014–2026
Bitcoin	crypto-asset, traded 24/7	2014–2026
EUR/RON	exchange rate, managed float	Jul 2005–2026
Gold	spot price, USD per ounce	1990–2026

- Daily closing values up to 18 September 2026 (repository folder data/market)
- Unless stated otherwise, returns are daily log returns (defined in Section 2)

The Five Series

- ▣ **S&P 500**: Standard & Poor's index of 500 large US companies; a **price index**, dividends are not included
- ▣ **BVB**: the Romanian stock exchange
 - ▶ **BET**: the benchmark index of the BVB
 - ▶ **BET-TR**: its version with dividends re-invested
- ▣ **EUR/RON**: price of one euro in Romanian lei; a **managed float** (market rate with central-bank interventions)
 - ▶ Source: the daily reference rate of the **BNR**
- ▣ **Gold**: spot price in US dollars (USD) per troy ounce; weekday quotes only (weekend prints removed)
- ▣ **Bitcoin**: price in USD; trades 24 hours a day, 7 days a week (24/7)

Data Quality Notes

- EUR/RON: official BNR reference rate instead of exchange quotes
 - ▶ Exchange quote series showed lag-1 autocorrelation (correlation of consecutive daily changes) of -0.4 to -0.7 in several years
 - ▶ Such values come from quote errors, not from the market
- S&P 500 **OHLC** data: Open, High, Low and Close prices of each day
 - ▶ Open prices before 2007 equal the previous close: unusable for range-based estimators
 - ▶ Range estimators therefore start in 2008 (Section 6)
- Lesson: always inspect the data before computing statistics

Prices Before Computers: the Stock Ticker



Stock ticker designed by Thomas Edison
(Smithsonian collection)

Photo: Jaclyn Nash, Smithsonian National Museum of
American History; public domain; Wikimedia Commons

- Before computers, trades reached brokers as prices printed on a paper **ticker tape**
- A daily close is one snapshot of this stream of prices; returns are built from such snapshots
- The choice of snapshot (close, fixing, a midnight snapshot) matters, as Chapter 0 showed
- Prices at the level of individual trades: market microstructure, Chapter 10

Prerequisites Recap: Returns

	Simple return R_t	Log return r_t
Definition	$(P_t + D_t)/P_{t-1} - 1$	$\ln(1 + R_t) = \ln P_t - \ln P_{t-1}$ (no dividend)
Over k periods	$\prod_j (1 + R_{t-j}) - 1$	$\sum_j r_{t-j}$: additive over time
Across assets	$R_{p,t} = \sum_j w_j R_{i,t}$: exact	only approximately
Range	$R_t \geq -100\%$	$(-\infty, +\infty)$
Use	portfolios, performance reporting	single-asset dynamics, CLT arguments

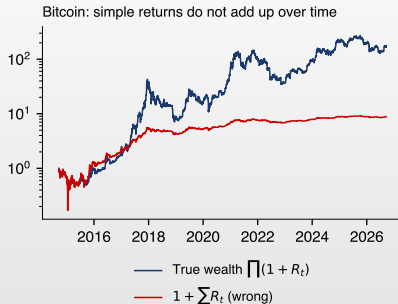
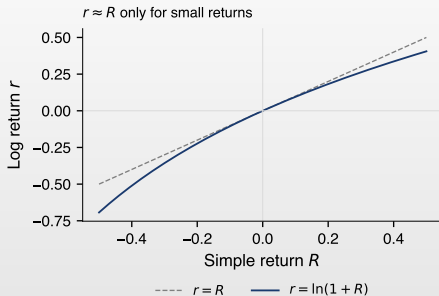
- ▣ Returns, not prices: returns are scale-free; log prices have a unit root, returns are close to stationary
- ▣ Dividends and splits: use adjusted or total-return series (BET-TR); P_t is the close, D_t the dividend paid in $(t-1, t]$
- ▣ $\ln(1 + R) < R$ for every $R \neq 0$: $+1\% \rightarrow 0.995\%$, but $-50\% \rightarrow -69.3\%$
- ▣ Textbooks: [Campbell, Lo & MacKinlay \(1997\)](#), ch. 1; [Franke, Härdle & Hafner \(2019\)](#)

Prerequisites Recap: Annualisation, CAGR and Volatility Drag

- P = periods per year: 252 (stocks), 365 (crypto, traded every day), the actual ≈ 260 weekday quotes (gold)
- $\mu_{\text{ann}} = P \bar{r}$ needs only a constant mean; $\sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$ is exact for serially uncorrelated returns with constant variance
 - ▶ Weakly stationary returns (constant mean and variance, autocovariances depending only on the lag):

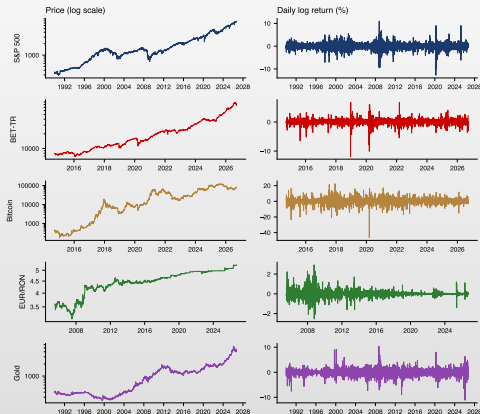
$$\text{Var}\left(\sum_{j=1}^h r_{t+j}\right) = h\sigma^2 \left[1 + 2 \sum_{j=1}^{h-1} (1 - j/h) \rho_j\right]$$
 - ▶ Autoregression of order 1 with $\rho_1 = 0.17$ (EUR/RON): factor $\sqrt{(1 + \rho_1)/(1 - \rho_1)} = 1.19$ for large h , annual volatility 4.4% $\rightarrow$ 5.3%; the variance-ratio test is in Chapter 2
- **CAGR** (Compound Annual Growth Rate) = $(P_T/P_0)^{1/Y} - 1$, Y = number of years; the geometric mean $\approx \mu_{\text{arith}} - \sigma^2/2$ (**volatility drag**)
 - ▶ S&P 500 1990–2026: $\hat{\sigma} = 1.133\%$ a day, 18.0% a year; CAGR 8.69% (price index, no dividends)
 - ▶ Bitcoin 2014–2026: arithmetic mean 65.2% a year ($P = 365$), CAGR 53.9%: a drag of 11 pp (percentage points)

Simple vs Log Returns in Practice



- Left: $r \approx R$ only for small moves; the gap grows with $|R|$
- Right: Bitcoin 2014–2026 multiplied wealth by 176.9, i.e. $\sum_t r_t = \ln 176.9 = 5.18$
- Adding daily simple returns gives $1 + \sum_t R_t = 8.83$: wrong by a factor of 20

Five Markets: Prices and Returns



- ▣ **Left:** prices on log scale, trending and non-stationary
- ▣ **Right:** daily log returns
 - ▶ Mean-reverting around zero
 - ▶ Calm and turbulent periods alternate
 - ▶ Extreme days cluster: 2008, March 2020
- ▣ EUR/RON moves more than 1% on only 1.4% of days
- ▣ Bitcoin, 12 March 2020: log return -46.5% , i.e. a price fall of $1 - e^{-0.465} = 37.2\%$

Recap: Prices and Returns

- Work with returns: scale-free and close to stationary
- Simple returns aggregate across assets; log returns aggregate over time
- Annualise with P and $\sqrt{P}$; use 365 for crypto
- CAGR (geometric) is below the arithmetic mean by about $\sigma^2/2$
- Use total-return or adjusted series when dividends matter

Sample Moments and Their Sampling Error

- ▣ $z_t = (r_t - \bar{r})/\hat{\sigma}$; skewness $\hat{S} = \frac{1}{T} \sum_t z_t^3$, kurtosis $\hat{K} = \frac{1}{T} \sum_t z_t^4$; Normal distribution: $S = 0$, $K = 3$, excess kurtosis $K - 3 = 0$
- ▣ Mean: $SE(\bar{r}) = \sigma/\sqrt{T}$; daily means are 10–60 times smaller than $\hat{\sigma}$, so the mean is the least precise moment
 - ▶ S&P 500: $\bar{r} = 0.033\% \pm 0.023\%$ a day (95%), $T = 9,245$
- ▣ i.i.d. data from the Normal distribution: $\sqrt{T} \hat{S} \rightarrow N(0, 6)$ and $\sqrt{T}(\hat{K} - 3) \rightarrow N(0, 24)$
- ▣ Any i.i.d. distribution: the delta method gives a Normal limit for $\hat{S}$ only if $\mathbb{E}r^6 < \infty$, for $\hat{K}$ only if $\mathbb{E}r^8 < \infty$
- ▣ **Tail index** α : $P(|X| > x) \approx c x^{-\alpha}$ for large x ; $\mathbb{E}|X|^p < \infty$ only for $p < \alpha$
 - ▶ With $\alpha \approx 3$ (Hill estimates, Section 4) $\hat{S}$ has no Normal limit and the population kurtosis does not exist

Three Classical Tests and What They Assume

Test	Statistic	H_0	χ^2 limit needs	Problem here
JB, Jarque & Bera (1980)	$T \left[\frac{\hat{S}^2}{6} + \frac{(\hat{K}-3)^2}{24} \right]$	i.i.d., Normal distribution	H_0 itself (all moments exist)	a rejection does not say which part fails
LB, Ljung & Box (1978)	$T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}$	$\rho_1 = \dots = \rho_m = 0$	i.i.d., not only uncorrelated	clustering: over-rejects
ARCH-LM, Engle (1982)	TR^2 , $\hat{\varepsilon}_t^2$ on q lags	i.i.d., no ARCH effects	$\mathbb{E}r^4 < \infty$	$\alpha \approx 3 < 4$: size unreliable

- ▣ **ARCH** = AutoRegressive Conditional Heteroskedasticity: today's variance depends on past returns; $\hat{\varepsilon}_t = r_t - \bar{r}$, $q = 5$
- ▣ We use $m = 10$; 5% critical values: $\chi_2^2 = 5.99$, $\chi_5^2 = 11.1$, $\chi_{10}^2 = 18.3$
- ▣ The question is not how to run the tests but whether their reference distribution holds for returns

Worked Example: JB for the S&P 500

- ▣ $JB = T \left[\hat{S}^2/6 + (\hat{K} - 3)^2/24 \right]$
- ▣ S&P 500, 1990–2026: $T = 9,245$, $\hat{S} = -0.364$, $\hat{K} - 3 = 10.88$
- ▣ Skewness term: $\hat{S}^2/6 = 0.132/6 = 0.022$
- ▣ Kurtosis term: $10.88^2/24 = 118.4/24 = 4.93$
- ▣ $JB = 9,245 \times (0.022 + 4.93) \approx 45,800$, against the critical value 5.99
- ▣ Almost all of the statistic comes from the kurtosis term: the problem is the tails, not the asymmetry

What JB Tests, and What a Rejection Means

- JB is a Lagrange Multiplier test [Jarque & Bera \(1980\)](#): under i.i.d. returns from the Normal distribution $\sqrt{T}(\hat{S}, \hat{K} - 3) \rightarrow N(0, \text{diag}(6, 24))$, hence χ^2_2
- Under this null all moments exist: the p-value is valid for the **joint** null “i.i.d. and Normal distribution”
 - ▶ Heavy tails (Hill $\hat{\alpha} \approx 3$) make JB grow with T : this is the power of the test, not a broken p-value
- A rejection does not say which part fails: the marginal distribution or independence (GARCH dependence changes $\text{Var}(\hat{S})$ and $\text{Var}(\hat{K})$)
 - ▶ [Bai & Ng \(2005\)](#): tests of skewness, kurtosis and normality of the marginal distribution of a time series, studentised with HAC variances
 - ▶ Their separate skewness and kurtosis tests (null $S = 0$ or $K = 3$ without normality) need $\mathbb{E}r^6 < \infty$ and $\mathbb{E}r^8 < \infty$
- Report as well: quantile-based skewness and kurtosis (next slide) and the tail index (Section 4)

Robust Alternative: Quantile-Based Skewness and Kurtosis

- Hinkley skewness $\frac{q_{0.95} + q_{0.05} - 2q_{0.5}}{q_{0.95} - q_{0.05}}$; Crow–Siddiqui kurtosis $\frac{q_{0.975} - q_{0.025}}{q_{0.75} - q_{0.25}} - 2.91$; both 0 for the Normal distribution [Kim & White \(2004\)](#)
- No moments needed, only positive denominators (interquantile ranges); 95% CI from a moving-block bootstrap (blocks of 20 days, 1,000 resamples), valid for stationary, weakly dependent returns with a positive density at these quantiles

	Moment skew.	Hinkley [95% CI]	Excess kurt.	Crow–Siddiqui [95% CI]
S&P 500	−0.36	−0.064 [−0.092, −0.041]	10.9	1.54 [1.29, 1.79]
BET-TR	−1.40	−0.036 [−0.085, 0.025]	17.8	1.06 [0.73, 1.43]
Bitcoin	−0.70	−0.018 [−0.069, 0.032]	12.0	2.25 [1.88, 2.64]
EUR/RON	0.73	0.041 [−0.022, 0.093]	15.1	4.87 [3.74, 6.13]
Gold	−0.23	0.004 [−0.025, 0.030]	9.3	1.47 [1.29, 1.73]

- Excess quantile kurtosis survives in all five markets: no CI contains 0
- Negative skewness survives only for the S&P 500: BET-TR's −1.40 comes from a few days (19 December 2018)

The ACF Under Conditional Heteroskedasticity

- ▣ **ACF** (AutoCorrelation Function): $\hat{\rho}_k = \sum_t x_t x_{t-k} / \sum_t x_t^2$, $x_t = r_t - \bar{r}$, plotted against k
- ▣ Martingale difference (uncorrelated, but $\mathbb{E}[r_t^2 \mid \text{past}]$ varies): $\hat{\rho}_k \approx N(0, \tau_k)$, $\hat{\tau}_k = \sum_t x_t^2 x_{t-k}^2 / (\sum_t x_t^2)^2$ [Romano & Thombs \(1996\)](#)
 - ▶ i.i.d.: $\tau_k = 1/T$ and the band $\pm 1.96/\sqrt{T}$; volatility clustering: $T\tau_k > 1$, the i.i.d. band is too narrow

Lag 1	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
$\hat{\rho}_1$	-0.079	0.061	-0.022	0.170	-0.011
$T\hat{\tau}_1$	5.00	4.35	2.89	6.85	2.43
t-statistic, i.i.d. / robust	-7.6 / -3.4	3.3 / 1.6	-1.5 / -0.9	12.4 / 4.7	-1.1 / -0.7

- ▣ Robust portmanteau $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k$ [Lobato, Nankervis & Savin \(2001\)](#): S&P 500 $Q(10) = 91.6$ but $\tilde{Q}(10) = 19.0$ ($p = 0.041$)
 - ▶ Needs $\mathbb{E}r^4 < \infty$ and $\mathbb{E}[x_t^2 x_{t-k} x_{t-l}] = 0$ for $k \neq l$; [Dalla, Giraitis & Phillips \(2022\)](#) keep only significant cross terms: none for the S&P 500
- ▣ ACF of $|r_t|$ and r_t^2 : Normal limits need $\alpha > 4$ and $\alpha > 8$ ([Davis & Mikosch \(1998\)](#), Section 4)

Summary Statistics: Five Markets

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Observations	9245	2992	4384	5352	9559
Mean (% per day)	0.033	0.077	0.118	0.007	0.025
Std. dev. (% per day)	1.13	0.98	3.49	0.28	1.01
Annualised volatility (%)	18.0	15.5	66.6	4.4	16.3
Skewness	-0.36	-1.40	-0.70	0.73	-0.23
Excess kurtosis	10.9	17.8	12.0	15.1	9.3
Min / max log return (%)	-12.8 / 11.0	-11.9 / 6.7	-46.5 / 22.5	-2.5 / 2.9	-11.1 / 10.4
Jarque-Bera	45805	40433	26482	51575	34732
Ljung-Box $Q(10), r_t$	91.6	70.0	20.2	221.7	32.1
Ljung-Box $Q(10), r_t^2$	8018	714	213	2420	1049
ARCH-LM (5 lags)	2229	371	115	871	483

- Min and max are daily **log** returns; EUR/RON from 1 July 2005 (new leu, RON)
- 5% critical values: $\chi^2_2 = 5.99$, $\chi^2_{10} = 18.3$, $\chi^2_5 = 11.1$; all i.i.d. p-values < 0.03, valid only under the assumptions above

Reading the Table

- ▣ **“i.i.d. and Normal distribution” is rejected everywhere:** JB in the tens of thousands; the quantile-based kurtosis shows that the tails are part of the reason
- ▣ **Heavy tails:** excess kurtosis between 9.3 (gold) and 17.8 (BET-TR)
- ▣ **Negative skewness** for equity, crypto and gold; strongest for BET-TR (-1.40)
- ▣ **Volatility is predictable:** $Q(10)$ on r_t^2 is 10–90 times larger than on r_t ; ARCH-LM rejects everywhere
- ▣ **Linear autocorrelation** of r_t : significant with i.i.d. tests, economically small
 - ▶ With robust standard errors only the S&P 500 (lag 1, $t = -3.4$) and EUR/RON ($t = 4.7$) remain significant
- ▣ EUR/RON is special: tiny volatility, high excess kurtosis, positive skewness, strong autocorrelation (Section 5)

Recap: Descriptive Statistics and Tests

- ▣ Skewness and kurtosis have Normal limits only if $\mathbb{E}r^6$ and $\mathbb{E}r^8$ are finite
- ▣ JB tests the joint null “i.i.d. and Normal distribution”; Ljung–Box and ARCH-LM have χ^2 limits only under i.i.d.-type assumptions that returns violate
- ▣ Robust versions: quantile-based moments, heteroskedasticity-robust ACF bands and $\tilde{Q}$
- ▣ Heavy tails survive every robust check; linear predictability mostly does not
- ▣ Statistical significance is not economic significance in samples of thousands of days

The People Behind the Stylised Facts (1/2)



Carl Friedrich Gauss (1777–1855)

Painting: C.A. Jensen (1840); public domain;
Wikimedia Commons

- The Normal distribution: the benchmark that returns violate



Louis Bachelier (1870–1946)

Photo: unknown author; public domain;
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- Prices as a random walk, in his thesis Bachelier (1900)

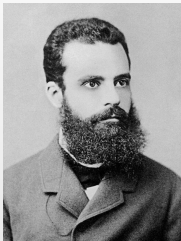


William S. Gosset, “Student” (1876–1937)

Photo: unknown author; public domain;
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- The Student- t distribution Student (1908), today a model for heavy tails

The People Behind the Stylised Facts (2/2)



Vilfredo Pareto (1848–1923)

Photo: unknown author (1870s); public domain; [Wikimedia Commons](#)



Benoît Mandelbrot (1924–2010)

Photo: Rama (2007); CC BY-SA 2.0 France; [Wikimedia Commons](#)

- Pareto: **power laws** for large incomes, the ancestor of heavy-tail models
- Mandelbrot: cotton prices have heavy tails and the Normal-distribution model fails [Mandelbrot \(1963\)](#)
- Their tail index α is what the Hill estimator in this chapter measures

The Stylised Facts of Cont (2001)

1. Absence of linear autocorrelation
 2. Heavy tails
 3. Gain/loss asymmetry
 4. Aggregational Gaussianity
 5. Intermittency
 6. Volatility clustering
 7. Conditional heavy tails
 8. Slow decay of autocorrelation in absolute returns
 9. Leverage effect
 10. Volume/volatility correlation
 11. Asymmetry in time scales
- ▣ We verify most of them on the five markets, with data up to September 2026
 - ▣ Early evidence: Mandelbrot (1963), Fama (1965); textbook treatment: Taylor (2008), Franke, Härdle & Hafner (2019)

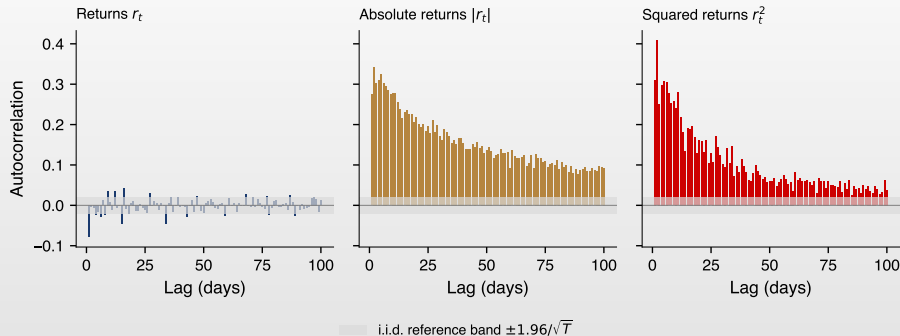
Do Cont's Facts Still Hold? Recent Evidence

- ▣ Ratliff-Crain, Van Oort, Koehler & Tivnan (2025)
 - ▶ Re-test all 11 facts on intraday returns of the Dow 30 stocks, October 2018 – March 2019
 - ▶ Conclusive evidence for 8 facts, no support for the remaining 3: a fact is a hypothesis to re-test on each market and frequency
- ▣ Heavy tails beyond the Student- t : estimation of alpha-stable distributions, whose tail index is below 2 [Pele \(2014\)](#)
- ▣ Entropy-based measures of tail risk as an alternative to kurtosis [Pele, Lazar & Dufour \(2017\)](#)
- ▣ Crypto-assets: the same facts, with a tail factor that separates them from other asset classes [Pele et al. \(2023\)](#)
- ▣ This chapter re-tests the facts on five markets and adds the inference that each test needs

Facts 1 and 6: Uncorrelated but Not Independent

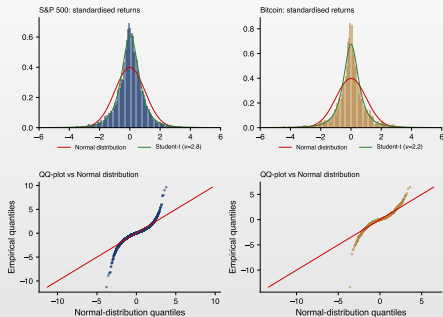
- **Fact 1 — absence of autocorrelation:** $\text{corr}(r_t, r_{t-k}) \approx 0$ for $k \geq 1$
 - ▶ Past returns do not predict the direction of future returns linearly
- **Fact 6 — volatility clustering:** large moves tend to be followed by large moves, of either sign
 - ▶ $\text{corr}(|r_t|, |r_{t-k}|) > 0$ for many lags
- Together: returns are **uncorrelated but not independent**
- Independence would require zero correlation of **every** function of returns, including $|r_t|$
- Modelling: $r_t = \sigma_t \varepsilon_t$ with σ_t driven by past returns (GARCH, Chapter 5); linear autoregressive forecasts of r_t add little

Facts 1 and 6 on the S&P 500



- S&P 500 1990–2026, lags 1–100; grey band: i.i.d. $\pm 1.96/\sqrt{T}$; the robust band is 2.2 times wider at lag 1
- Returns: almost all autocorrelations inside the band (lag 1: -0.08)
- Absolute and squared returns: large, positive and persistent (lag 1: 0.27 for $|r_t|$)

Heavy Tails: Histograms and QQ-Plots



- Fact 2: extreme returns are far more frequent than under a Normal distribution with the same variance
- Standardised returns $(r_t - \bar{r})/\hat{\sigma}$: higher peak and fatter tails than the Normal distribution
- A Student- t fits much better (next slides); QQ-plots (empirical quantiles against those of the Normal distribution) bend away from the line in both tails

How Heavy Are the Tails?

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Days beyond $3\hat{\sigma}$ (%)	1.55	1.44	1.82	2.06	1.42
Days beyond $5\hat{\sigma}$ (count)	30	12	10	29	26
Expected under the Normal distribution, $> 5\hat{\sigma}$	0.005	0.002	0.003	0.003	0.005
Student- t degrees of freedom $\hat{\nu}$ (maximum likelihood)	2.8	3.1	2.2	1.2	2.9

- Normal distribution: 0.27% of days beyond 3σ ; observed 5–8 times more
- Normal distribution: one 5σ day every 6,900 years of trading; observed 10–30 in 12–36 years
- $\hat{\nu}$: Student- t degrees of freedom (next slide), estimated by **maximum likelihood**: the value that makes the observed returns most probable

The Student- t Distribution: Tails, Moments and Our Fits

- Density with ν degrees of freedom, location μ , scale s : $f(x) \propto \left(1 + \frac{(x-\mu)^2}{\nu s^2}\right)^{-(\nu+1)/2}$; $\nu \rightarrow \infty$: the Normal distribution; $\nu = 1$: Cauchy; for returns: [Praetz \(1972\)](#)
- Power-law tail $P(X > x) \propto x^{-\nu}$: moments exist only of order $< \nu$
 - ▶ Variance $s^2\nu/(\nu - 2)$ for $\nu > 2$; excess kurtosis $6/(\nu - 4)$ for $\nu > 4$
- Our fits $\hat{\nu} \approx 1.2\text{--}3.1$ are **model-based** tail indices: valid if the Student- t holds everywhere; the many small returns in the centre dominate the likelihood, so a misfit centre moves $\hat{\nu}$
- $\hat{\nu} \leq 2$ (EUR/RON: 1.2) implies infinite variance; a finite sample always has a finite variance, so this alone is no contradiction
 - ▶ But Hill (tail only) gives 2.8: the disagreement signals misspecification (the fixing: 57% of daily changes below 0.1%, plus rare jumps)
- The tail itself: the Hill estimator (next slides); power laws were first argued by [Mandelbrot \(1963\)](#) and [Fama \(1965\)](#)

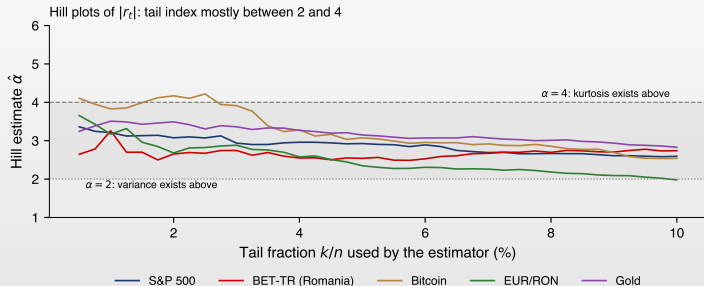
The Hill Estimator of the Tail Index

- ▣ **Tail index** α : $P(|X| > x) \approx c x^{-\alpha}$ for large x ; moments of order $< \alpha$ exist
 - ▶ $\alpha > 2$: finite variance; $\alpha > 4$: finite kurtosis
- ▣ **Hill estimator**, [Hill \(1975\)](#): order the absolute returns $X_{(1)} \geq X_{(2)} \geq \dots \geq X_{(n)}$ and use the k largest

$$\hat{\alpha}_k = \left[\frac{1}{k} \sum_{i=1}^k \ln X_{(i)} - \ln X_{(k+1)} \right]^{-1}$$

- ▣ **Choice of k** : a bias–variance trade-off
 - ▶ Small k : few observations, noisy; large k : the centre of the distribution enters, biased
 - ▶ **Hill plot**: $\hat{\alpha}_k$ against k ; read the level where the curve is flat (here $k = 1\text{--}5\%$ of n)

Hill Plots for Five Markets



- ▣ $\hat{\alpha}$ for $|r_t|$ at $k = 2.5\%$ of n : S&P 500 3.1, BET-TR 2.7, Bitcoin 4.2, EUR/RON 2.8, gold 3.3
- ▣ For $k = 1\text{--}5\%$: $\hat{\alpha}$ between 2.3 and 3.5 except Bitcoin (3.1–4.2); loss tails heavier for BET-TR (2.2 vs 3.1) and Bitcoin (2.7 vs 3.4)
- ▣ Point estimates above 2 suggest a finite variance (with the bootstrap SE, BET-TR's interval reaches below 2); except possibly for Bitcoin, the **fourth moment very likely does not exist**: sample kurtosis never settles (confidence intervals: next slides); tails are modelled with extreme value theory (Chapter 7)

Hill Estimator: Asymptotics and the Choice of k

- ▣ Hill = maximum likelihood estimator of a Pareto tail above $u = X_{(k+1)}$, conditional on u (Seminar A11)
- ▣ If $k \rightarrow \infty$ and $k/n \rightarrow 0$, and the tail is Pareto up to a second-order term

$$\sqrt{k}(\hat{\alpha}_k - \alpha) \rightarrow N(0, \alpha^2), \quad \text{SE}(\hat{\alpha}_k) \approx \hat{\alpha}_k / \sqrt{k}$$

- ▶ If k grows too fast, a bias term appears: the tail is only approximately Pareto
- ▣ Dependent data (GARCH-type): $\hat{\alpha}$ stays consistent, but extremes cluster and the variance exceeds α^2/k [Hsing \(1991\)](#)
- ▣ Choosing k : plot $\hat{\alpha}$ against $\log k$ ("alt-Hill" plot) [Drees, Resnick & de Haan \(2000\)](#)
- ▣ Regression alternative: log rank on log size, with rank shifted by $-\frac{1}{2}$; $\text{SE} = \hat{\alpha}\sqrt{2/k}$ [Gabaix & Ibragimov \(2011\)](#)

Tail Indices with Confidence Intervals

$ r_t $, $k = 2.5\%$ of n	n	k	$\hat{\alpha}$	SE i.i.d.	95% CI (i.i.d.)	SE block bootstrap
S&P 500	9245	231	3.07	0.20	[2.68, 3.47]	0.33
BET-TR	2992	74	2.67	0.31	[2.06, 3.28]	0.45
Bitcoin	4384	109	4.22	0.40	[3.43, 5.01]	0.47
EUR/RON	5352	133	2.82	0.24	[2.34, 3.30]	0.28
Gold	9559	238	3.31	0.21	[2.89, 3.73]	0.24

- Moving-block bootstrap (blocks of 20 days, 500 resamples) keeps the clustering of extremes: SEs 1.1–1.6 times the i.i.d. ones
- $\alpha = 4$ is outside every interval except Bitcoin's, also with the bootstrap SE: the fourth moment very likely does not exist
- $\alpha = 2$ is outside every i.i.d. interval; with the bootstrap SE only BET-TR reaches it ($2.67 - 1.96 \times 0.45 = 1.79$)

Two Tails and the Tail of $|r_t|$

- $P(|X| > x) = P(X > x) + P(X < -x)$: with indices α_- (losses) and α_+ (gains), $|X|$ has index $\min(\alpha_-, \alpha_+)$
- At the same k the estimates are not comparable: the k largest $|r_t|$ come from both tails, so their threshold lies further out
- Common threshold u : the lower of the two one-sided thresholds at $k = 2.5\%$ of n

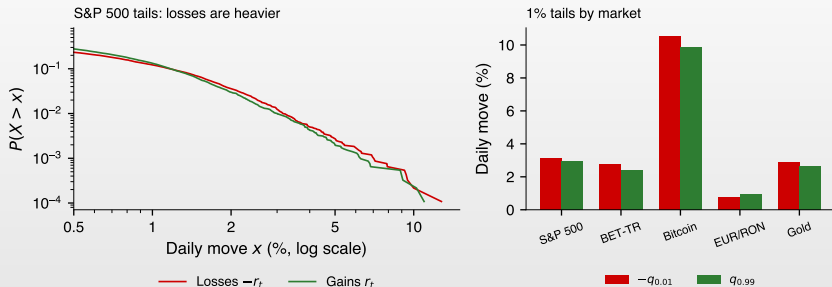
	u (%)	k for $ r_t $	$\hat{\alpha}$, $ r_t $	$\hat{\alpha}_-$, losses	$\hat{\alpha}_+$, gains
S&P 500	2.17	508	2.90	2.85	2.96
BET-TR	1.79	153	2.54	2.20	3.06
Bitcoin	7.11	233	3.02	2.71	3.36
EUR/RON	0.54	295	2.28	2.63	2.04
Gold	1.98	513	3.11	2.97	3.28

- At a common u , the $|r_t|$ estimate lies between the two tails
- Bitcoin's 4.2 at $k = 2.5\%$ uses a higher threshold, where its Hill plot is not yet flat: second-order bias; report both one-sided indices

Why Heavy Tails Matter for Modelling

- ▣ **VaR** = Value-at-Risk: the loss exceeded with a small probability (e.g. 1%) over a horizon
 - ▶ It is a quantile of the loss distribution (Chapter 7)
- ▣ A Normal-distribution model with the right variance **underestimates** extreme quantiles
- ▣ Stress scenarios based on “ 3σ events” are far too optimistic
- ▣ Estimators based on high moments (kurtosis) are unstable: prefer quantiles and tail counts
- ▣ Remedies: heavy-tailed distributions (Student- t , possibly skewed), extreme value theory (Chapter 7)

Fact 3: Gain/Loss Asymmetry in the Data

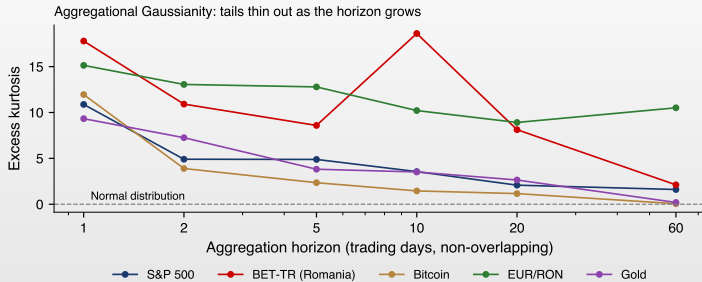


- **Fact 3:** large losses are more frequent than gains of the same size; survival functions $P(X > x)$ of losses $(-r_t)$ and gains (r_t) , log-log scale
- VaR 1% ($-q_{0.01}$, the size of the 1% return quantile) exceeds the 99% return quantile $q_{0.99}$ for 4 of 5 markets
 - S&P 500: 3.15% vs 2.97%; BET-TR: 2.73% vs 2.40%; Bitcoin: 10.5% vs 9.9%
- EUR/RON is the exception: leu depreciations (up-moves) are larger (0.94% vs 0.76%)

Fact 4: Aggregational Gaussianity

- **Statement:** as the horizon grows, returns get closer and closer to the Normal distribution
- h -day log return: $r_t(h) = r_t + r_{t-1} + \cdots + r_{t-h+1}$
 - ▶ We use non-overlapping blocks, so that the h -day returns do not share days
- Mechanism: the CLT applies to sums of many daily returns
- Convergence is slow because daily returns are dependent (volatility clustering)

Aggregational Gaussianity in the Data



- Excess kurtosis of h -day returns, $h = 1, \dots, 60$
- Tails thin out: S&P 500 from 10.9 (daily) to 1.6 ($h = 60$), Bitcoin from 12.0 to 0.1
- $h = 60$ is a quarter for the S&P 500 (252 days a year) but two months for Bitcoin (365 days a year)
- At $h = 60$ only 154 (S&P 500) and 73 (Bitcoin) non-overlapping returns remain: approximate standard error $\sqrt{24/n} = 0.39$ and 0.57 under normality, so these values are very imprecise

Why Heavy Tails? Mixtures of Normal Distributions

- ▣ Suppose $r_t = \sigma_t \varepsilon_t$ with $\varepsilon_t \sim N(0, 1)$ independent of a random volatility σ_t

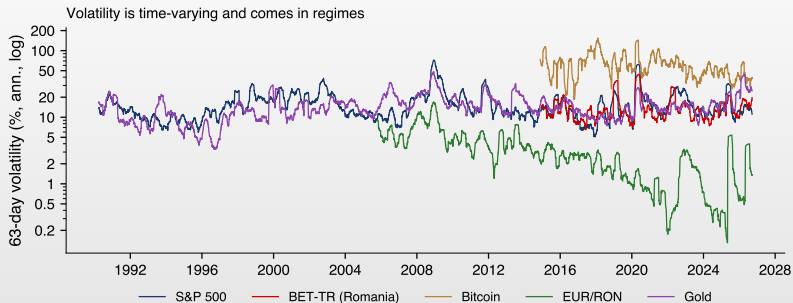
$$K = \frac{\mathbb{E}[r_t^4]}{\mathbb{E}[r_t^2]^2} = 3 \frac{\mathbb{E}[\sigma_t^4]}{\mathbb{E}[\sigma_t^2]^2} \geq 3$$

- ▣ Jensen's inequality ($\mathbb{E}[X^2] \geq \mathbb{E}[X]^2$) with $X = \sigma^2$: any variation in volatility creates excess kurtosis
- ▣ Example: $\sigma = 1\%$ on 90% of days and 3% on 10% of days

$$K = 3 \cdot \frac{0.9 \cdot 1 + 0.1 \cdot 81}{(0.9 + 0.1 \cdot 9)^2} = 3 \cdot \frac{9.0}{3.24} = 8.33$$

- ▣ This links heavy tails (Fact 2) to volatility clustering (Fact 6)

Fact 5: Intermittency in the Data

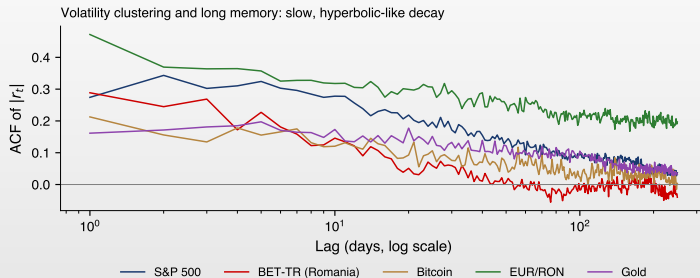


- Fact 5: bursts of high volatility separated by calm periods; constant-volatility models cannot produce them
- 63-day rolling volatility $\sqrt{P} \times$ the standard deviation over the window, log scale
- Common spikes across markets: 2008–2009, March 2020
- EUR/RON volatility collapsed after 2012: an exchange-rate regime managed by the central bank

Fact 8: Slow Decay — Long Memory in Volatility

- ▣ **Statement:** the ACF of $|r_t|$ stays positive for months
- ▣ **Short memory:** autocorrelations are absolutely summable, $\sum_k |\rho_k| < \infty$; a common case is **exponential decay**, $\rho_k \approx c \phi^k$ with $|\phi| < 1$
 - ▶ Example: GARCH(1,1) volatility (Chapter 5)
- ▣ **Long memory:** autocorrelations decay **hyperbolically**, $\rho_k \approx c k^{2d-1}$, memory parameter $0 < d < \frac{1}{2}$
 - ▶ Shocks to volatility die out very slowly
- ▣ Evidence: [Ding, Granger & Engle \(1993\)](#)

Long Memory in the Data



- ACF of $|r_t|$ up to one year (250 lags), log scale on the lag axis
- S&P 500: 0.27 at lag 1, 0.22 at lag 20, still 0.09 at lag 100
- BET-TR decays fastest (shorter sample); decay is slower than the exponential decay of GARCH(1,1)

Estimating the Memory Parameter d

- Long memory: spectral density $f(\lambda) \approx G\lambda^{-2d}$ as $\lambda \rightarrow 0$, $0 < d < \frac{1}{2}$; semiparametric estimators use only the first m Fourier frequencies $\lambda_j = 2\pi j/T$ and the periodogram $I(\lambda_j)$
- Log-periodogram regression: $\ln I(\lambda_j)$ on $-2 \ln(2 \sin(\lambda_j/2))$, slope $\hat{d}$, $\text{SE} = \pi/\sqrt{24m}$ Geweke & Porter-Hudak (1983)
- Local Whittle: $\hat{d} = \arg \min_d [\ln(\frac{1}{m} \sum_j \lambda_j^{2d} I(\lambda_j)) - \frac{2d}{m} \sum_j \ln \lambda_j]$, $\sqrt{m}(\hat{d} - d) \rightarrow N(0, \frac{1}{4})$, robust to non-Gaussianity Robinson (1995)

$ r_t $, $m = T^{0.65}$	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
m	378	181	232	265	386
Local Whittle $\hat{d}$ (SE)	0.48 (0.03)	0.37 (0.04)	0.35 (0.03)	0.39 (0.03)	0.40 (0.03)
Log-periodogram $\hat{d}$ (SE)	0.50 (0.03)	0.41 (0.05)	0.35 (0.04)	0.38 (0.04)	0.42 (0.03)

- $\hat{d} \approx 0.35$ – 0.50 everywhere, at least 8 SEs from $d = 0$ (short memory) with both estimators; the S&P 500 is close to the stationarity boundary $\frac{1}{2}$

Long Memory or Structural Breaks?

- Rare shifts in the level of volatility produce a slowly decaying ACF of $|r_t|$ and $\hat{d} > 0$ without long memory Diebold & Inoue (2001)
- Changes in the unconditional variance mimic long memory and integrated GARCH Mikosch & Stărică (2004)
- Check: locate one variance break with the cumulative sum of squares Inclán & Tiao (1994), then re-estimate d (local Whittle) on each side

$ r_t $	Break date	Volatility before / after (%)	$\hat{d}$ before (SE)	$\hat{d}$ after (SE)
S&P 500	20 Jul 1998	12.6 / 19.3	0.28 (0.04)	0.51 (0.03)
BET-TR	17 Dec 2018	12.5 / 16.9	0.19 (0.05)	0.37 (0.04)
Bitcoin	19 Jun 2022	74.9 / 47.7	0.39 (0.04)	0.15 (0.05)
EUR/RON	27 Jun 2013	6.6 / 2.2	0.35 (0.04)	0.35 (0.04)
Gold	29 Nov 2005	13.2 / 18.3	0.30 (0.03)	0.37 (0.03)

- $\hat{d}$ stays well above 0 in every sub-sample: one break does not explain the memory
- But $\hat{d}$ moves (Bitcoin $0.39 \rightarrow 0.15$): several breaks or a time-varying d remain open; the break dates are descriptive

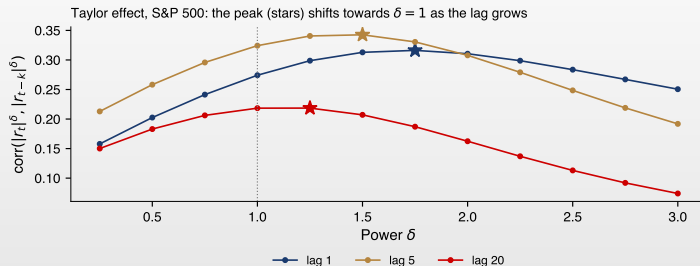
Infinite Fourth Moments and Statistics on r_t^2

- ▣ The ACF of r_t^2 is a ratio of fourth-order moments: it needs $E[r_t^4] < \infty$, i.e. tail index $\alpha > 4$
- ▣ Our Hill estimates are about 2.5–3.5: the fourth moment is very likely infinite
 - ▶ The sample ACF of r_t^2 then converges slowly, to a limit that is not the Normal distribution; the $\pm 1.96/\sqrt{T}$ band is not valid, [Davis & Mikosch \(1998\)](#)
 - ▶ Ljung–Box on r_t^2 and ARCH-LM inherit the problem: the statistics are huge, but their p-values are not reliable
- ▣ The conclusion “volatility clusters” is robust; the exact magnitudes are not
- ▣ Remedy: use $|r_t|$, whose ACF needs only $\alpha > 2$ — this is one reason behind the Taylor effect (next slide)

The Taylor Effect

- **Definition:** compute the autocorrelation of $|r_t|^\delta$ for different powers $\delta > 0$
 - ▶ $\delta = 1$: absolute returns; $\delta = 2$: squared returns
- **Taylor effect:** the autocorrelation is highest for δ around 1, not for $\delta = 2$
- Implication: absolute returns are a more persistent, more robust measure of volatility than squared returns
- Documented by Taylor (1986); the name “Taylor effect” was given by Granger & Ding (1995); see also Ding, Granger & Engle (1993)

The Taylor Effect in the Data



- S&P 500: autocorrelation of $|r_t|^\delta$ as a function of δ
- Maximum at $\delta = 1.75$ (lag 1), 1.5 (lag 5) and 1.25 (lag 20): the optimum moves towards 1 at longer lags
- Motivates power-GARCH models and absolute-value volatility measures

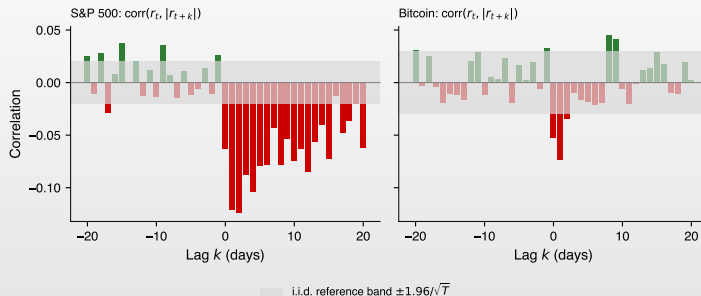
Fact 9: The Leverage Effect

- ▣ **Statement:** falling prices raise **future** volatility more than rising prices
- ▣ **Leverage correlation** at lag k

$$L(k) = \text{corr}(r_t, |r_{t+k}|)$$

- ▶ $L(k) < 0$ for $k > 0$: a negative return today is followed by larger moves
- ▶ $L(k) \approx 0$ for $k < 0$: past volatility does not predict today's sign
- ▣ Explanations: financial leverage of firms, [Christie \(1982\)](#); the retarded volatility model, [Bouchaud, Matacz & Potters \(2001\)](#)

The Leverage Effect in the Data

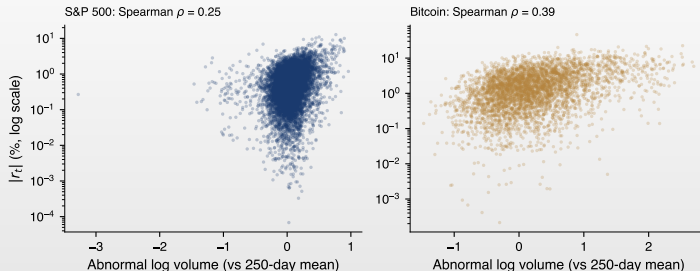


- S&P 500: $L(1) = -0.12$, still -0.07 at $k = 10$; no effect for $k < 0$
- BET-TR -0.12 ; weak for Bitcoin (-0.07), negligible for gold (-0.02), reversed for EUR/RON ($+0.10$)
- Modelling: asymmetric volatility models, GJR-GARCH (Glosten–Jagannathan–Runkle) and EGARCH (Exponential GARCH), Chapter 5

Fact 10: Volume and Volatility Move Together

- ▣ **Statement:** days with large price moves are days with high trading volume
- ▣ **Abnormal log volume:** $\ln V_t$ minus its 250-day moving average
- ▣ Measure: **Spearman rank correlation**, the correlation of ranks, robust to outliers and monotone non-linearity
- ▣ Mixture-of-distributions hypothesis, [Clark \(1973\)](#): prices move in a random “business time” driven by information flow, which also drives volume
- ▣ Survey of the price–volume relation: [Karpoff \(1987\)](#)

Volume and Volatility in the Data



- Abnormal log volume against $|r_t|$
- Spearman rank correlation: 0.25 for the S&P 500 and 0.39 for Bitcoin
- Volume is a useful additional input for volatility forecasts
- Bitcoin caveat: volume is exchange-reported and aggregated over many venues; part of it is **wash trading** (an exchange trading with itself to inflate volume), [Cong, Li, Tang & Yang \(2023\)](#)

Facts 7 and 11: Conditional Tails and Time Scales

Fact 7: conditional heavy tails

- After removing volatility clustering, standardised returns are still heavy-tailed
- Fit a volatility model, e.g. GARCH (Chapter 5), get $\hat{\sigma}_t$, compute $\hat{z}_t = r_t / \hat{\sigma}_t$
- Kurtosis of $\hat{z}_t$: lower than for r_t , still above 3
- Models need heavy-tailed innovations (Student- t , possibly skewed)

□ Both facts are documented in [Cont \(2001\)](#)

□ Returns standardised by realised volatility from intraday data are close to the Normal distribution [Andersen, Bollerslev, Diebold & Ebens \(2001\)](#) (Chapter 9)

Fact 11: asymmetry in time scales

- Weekly volatility predicts daily volatility better than the reverse
- Traders with different horizons react at different frequencies
- **HAR-RV**: Heterogeneous AutoRegressive model of Realised Volatility (Chapter 9)

Recap: The Stylised Facts

- ▣ Uncorrelated returns, strongly dependent volatility (Facts 1, 6, 8)
- ▣ Heavy tails, also conditionally, that thin out slowly with aggregation (Facts 2, 4, 7)
- ▣ Crashes sharper than rallies in equity markets (Facts 3, 9)
- ▣ Volatility in regimes, linked to volume and information flow (Facts 5, 10)
- ▣ Different horizons interact asymmetrically (Fact 11)

Crypto-Assets: The Same Facts, Amplified

Bitcoin vs S&P 500

- ▣ Volatility 66.6% vs 18.0% per year
- ▣ Excess kurtosis 12.0 vs 10.9
- ▣ Skewness -0.70 vs -0.36
- ▣ Worst day, log return: -46.5% (12 March 2020) vs -12.8% (16 March 2020)
- ▣ As price changes: -37.2% vs -12.0%

But some facts are weaker

- ▣ Leverage effect: weak, short-lived
- ▣ Volatility memory: ACF of $|r_t|$ decays faster (lag 100: 0.02 vs 0.09)
- ▣ Trading 24/7: 365 periods per year

Crypto-Assets: An Asset Class of Their Own

- Crypto as an asset class with its own statistical profile: [Härdle, Harvey & Reule \(2020\)](#)
- Tail, memory and moment factors classify cryptos as a separate asset class, mainly through the tail factor: [Pele et al. \(2023\)](#)
- Crypto modelling and the CRIX index in quantitative finance: [Härdle, Chen & Overbeck \(2017\)](#)
 - ▶ CRIX = CRypto IndeX, a benchmark index of the crypto market
- Deeper treatment in Chapter 16 (digital assets)

EUR/RON: A Managed Exchange Rate

- ▣ **Managed float:** the exchange rate is market-determined, but the central bank intervenes to limit fluctuations
- ▣ BNR reference rate, 1 July 2005–2026: from the new leu (RON, 1 RON = 10,000 old lei), as in Chapter 0
 - ▶ **Low volatility** (4.4% per year) but **high excess kurtosis** (15.1)
 - ▶ Long calm periods interrupted by sudden adjustments
 - ▶ Largest daily moves: October 2008 (+2.9% on 3 October, −2.5% on 9 October)
- ▣ **Positive** lag-1 autocorrelation (0.17), unlike equities
 - ▶ The central bank smooths adjustments over several days; a daily fixing, not a traded price
- ▣ Lessons: facts are market-specific; low volatility is not low tail risk; exchange quote series showed spurious negative autocorrelation (quote errors)

Romania: The BET Total Return Index

- ▣ BET-TR, 2014–2026, 2,992 days
- ▣ Facts confirmed: excess kurtosis 17.8, clustering ($Q(10)$ on r_t^2 : 714), VaR 1% 2.7%
- ▣ Strongest negative skewness of the five series (-1.40)
 - ▶ Worst day (log return) -11.9% on 19 December 2018: reaction to the bank-asset tax announced by the government on 18 December
 - ▶ The tax was adopted on 28 December 2018 as Government Emergency Ordinance (GEO) 114/2018
 - ▶ Policy shocks produce single-day crashes without comparable rallies

Romania: Autocorrelation in a Smaller Market

- ▣ Positive lag-1 autocorrelation (0.06), in contrast with the S&P 500 (-0.08)
- ▣ **Non-synchronous trading:** index constituents do not all trade at every moment
 - ▶ The index reflects news with a delay, creating positive autocorrelation, [Campbell, Lo & MacKinlay \(1997\)](#), ch. 3
- ▣ Frontier and emerging markets (below developed status in index-provider classifications)
 - ▶ Lower liquidity (it is costlier to trade quickly)
 - ▶ Stronger autocorrelation and policy-driven jumps

Why Use Open, High, Low and Close?

- **Close-to-close** estimator: the standard deviation of daily close-to-close log returns
 - ▶ Uses one number per day and ignores intraday information
- **Range-based** estimators use the OHLC prices
 - ▶ Notation: o_t, h_t, l_t, c_t are the log open, high, low and close of day t
 - ▶ The range $h_t - l_t$ is large on volatile days even if the close equals the open
- **Efficiency**: the ratio of variances of two estimators; higher efficiency = less noise for the same sample

Parkinson and Garman–Klass Estimators

- ▣ **Parkinson**, [Parkinson \(1980\)](#): uses only the daily range

$$\hat{\sigma}_P^2 = \frac{1}{4 \ln 2} \frac{1}{n} \sum_{t=1}^n (h_t - l_t)^2$$

- ▣ **Garman–Klass**, [Garman & Klass \(1980\)](#): combines the range with the open-to-close move

$$\hat{\sigma}_{GK}^2 = \frac{1}{n} \sum_{t=1}^n \left[\frac{1}{2} (h_t - l_t)^2 - (2 \ln 2 - 1) (c_t - o_t)^2 \right]$$

- ▣ Derived for a Brownian motion without drift and without overnight jumps
 - ▶ Brownian motion: the continuous-time random walk (Chapter 11)
 - ▶ Efficiency relative to close-to-close: about 5 (Parkinson) and 7–8 (Garman–Klass)

Rogers–Satchell and Yang–Zhang Estimators

- **Rogers–Satchell**, Rogers & Satchell (1991): unbiased under a non-zero **drift** (expected trend of the log price)

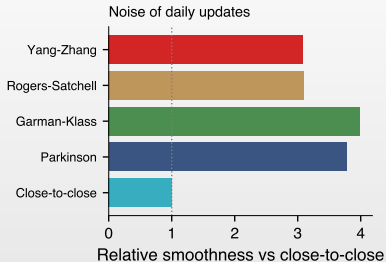
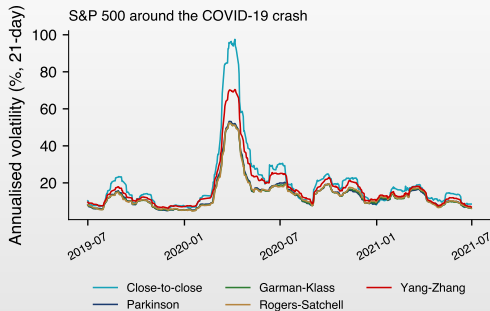
$$\hat{\sigma}_{RS}^2 = \frac{1}{n} \sum_{t=1}^n [(h_t - o_t)(h_t - c_t) + (l_t - o_t)(l_t - c_t)]$$

- **Yang–Zhang**, Yang & Zhang (2000): adds the overnight (close-to-open) jump

$$\hat{\sigma}_{YZ}^2 = \hat{\sigma}_{\text{overnight}}^2 + k \hat{\sigma}_{\text{open-close}}^2 + (1 - k) \hat{\sigma}_{RS}^2, \quad k = \frac{0.34}{1.34 + \frac{n+1}{n-1}}$$

- ▶ $\hat{\sigma}_{\text{overnight}}^2$: variance of $o_t - c_{t-1}$; $\hat{\sigma}_{\text{open-close}}^2$: variance of $c_t - o_t$
- ▶ Minimum-variance combination among drift-independent estimators robust to opening jumps
- Range estimators are the bridge to realised volatility from intraday data (Chapter 9)

Estimators on the S&P 500



- 21-day windows, 2008–2026; averages: close-to-close 16.4%, Yang–Zhang 13.9%, Parkinson 13.0%, Garman–Klass 12.4%, Rogers–Satchell 12.3%
- Pure range estimators are lower: they miss overnight gaps
- Yang–Zhang includes the overnight gap but stays 2.5 pp below close-to-close: the index open is partly stale (next slide)
- Range-based estimates are 3–4 times smoother (less noise in daily updates)

Data Quality: Stale Opens of the S&P 500 Index

- The index **open** is computed from the constituents' first prices; at 9:30 many stocks have not traded yet, so their previous close enters the open
- Open = previous close: about 96% of days until 2005, 3.5% over 2008–2026 (up to 26% in 2012–2013); on other days part of the open is still stale
- Comparison 2008–2026 with the SPY ETF, whose open is a traded price (annualised, %; the first two columns average 21-day estimates, the last two use the full sample)

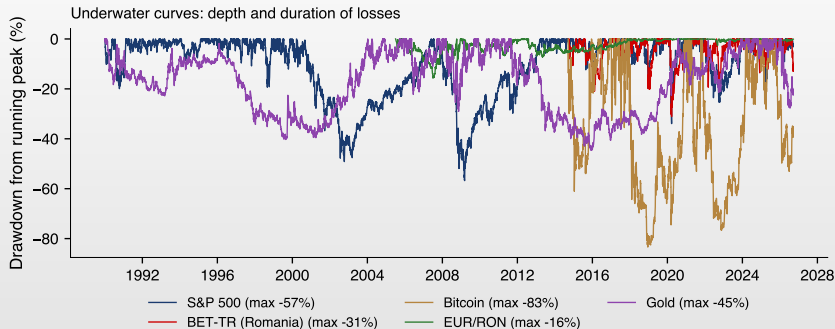
	Close-to-close	Yang–Zhang	Overnight	Open-to-close
S&P 500 index	16.4	13.9	6.7	17.3
SPY ETF	16.2	16.8	12.0	15.2

- A stale open moves part of the overnight move into the day: the index overnight volatility is understated and Yang–Zhang falls below close-to-close
- Lesson: for estimators that use the open (Garman–Klass, Rogers–Satchell, Yang–Zhang) use a traded instrument (ETF, futures), not the index

Drawdowns: Reminder and Focus

- Drawdown DD_t and maximum drawdown MDD were defined in Chapter 0
 - ▶ S&P 500: 9 October 2007 (1,565.15) to 9 March 2009 (676.53): -56.8% , which needs $+131\%$ to recover; peak regained on 28 March 2013
- New here: depth across five markets and **recovery duration**, the time from the peak until the price regains it
- MDD depends on the sample length: for a driftless Brownian motion of the log price, the maximum **log** drawdown $D = -\ln(1 + MDD) \geq 0$ has $\mathbb{E}[D] = \sqrt{\pi/2} \sigma \sqrt{T}$ Magdon-Ismail, Atiya, Pratap & Abu-Mostafa (2004)
 - ▶ $MDD = e^{-D} - 1$; drawdowns from samples of different length are not comparable (Seminar A4)
- Investors feel drawdowns more than volatility: a -50% loss needs $+100\%$ to recover

Drawdowns Across Markets



- Maximum drawdowns: Bitcoin -83%, S&P 500 -57%, gold -45%, BET-TR -31% (March 2020), EUR/RON -16%
- Depth and duration matter together

How Long Does Recovery Take?

	MDD	Peak	Trough	Peak regained
S&P 500	−57%	Oct 2007	Mar 2009	Mar 2013 (5.5 years)
BET-TR	−31%	Jan 2020	Mar 2020	Dec 2020 (0.9 years)
Bitcoin	−83%	Dec 2017	Dec 2018	Nov 2020 (3.0 years)
EUR/RON	−16%	Dec 2005	Jul 2007	Jan 2008 (2.0 years)
Gold	−45%	Sep 2011	Dec 2015	Jul 2020 (8.9 years)

- Depth and duration are different risks: gold's drawdown was shallower than Bitcoin's but lasted three times longer
- Fast V-shaped recoveries (BET-TR 2020) versus long underwater periods (gold 2011–2020)
- For EUR/RON a “drawdown” is a fall of the euro against the leu

Sharpe, Sortino and Calmar Ratios

- ▣ **Sharpe ratio** Sharpe (1966), Sharpe (1994): $SR = (\mu_{\text{ann}} - r_f) / \sigma_{\text{ann}}$ on **simple** returns, $\mu_{\text{ann}} = P\bar{R}$; $r_f = 0$ here, for comparability
 - ▶ S&P 500 1990–2026: $\mu_{\text{ann}} = 9.95\%$ (mean log return 8.33% plus $\approx \sigma^2/2 = 1.62$ pp), $\sigma_{\text{ann}} = 17.97\%$, $SR = 0.55$
- ▣ **Sortino ratio** Sortino & van der Meer (1991): $SoR = (\mu_{\text{ann}} - r_f) / \sigma_{\text{ann}}^-$, $\sigma^- = \sqrt{\frac{1}{T} \sum_{t=1}^T \min(R_t, 0)^2}$
 - ▶ The average runs over all T days, with positive days counted as 0; S&P 500: $\sigma_{\text{ann}}^- = 12.68\%$, $SoR = 0.78$
- ▣ **Calmar ratio** = $\text{CAGR} / |MDD|$; S&P 500: $8.69\% / 56.78\% = 0.15$, driven by one episode

Sharpe Ratios with Standard Errors

- Delta method, i.i.d. returns from the Normal distribution: $\text{Var}(\widehat{SR}_d) \approx (1 + SR_d^2/2)/T$ at the sampling frequency [Lo \(2002\)](#)
 - ▶ Annualised from T daily returns over Y years: $\text{SE}(\widehat{SR}_{\text{ann}}) \approx \sqrt{P(1 + SR_d^2/2)/T} \approx 1/\sqrt{Y}$: the span matters, not the frequency
 - ▶ i.i.d., any distribution with $\mathbb{E}r^4 < \infty$: $(1 - S SR_d + \frac{K-1}{4} SR_d^2)/T$ [Opdyke \(2007\)](#); here the SEs change by at most 0.02

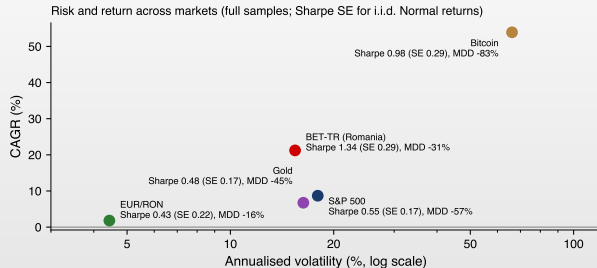
	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Years Y	36.7	12.0	12.0	21.2	36.7
Sharpe ratio (SE, Normal)	0.55 (0.17)	1.34 (0.29)	0.98 (0.29)	0.43 (0.22)	0.48 (0.17)

- Clustering: HAC or studentised block-bootstrap tests [Ledoit & Wolf \(2008\)](#); they still need finite fourth moments, which the Hill estimates put in doubt
 - ▶ BET-TR vs S&P 500, common days 2014–2026: difference 0.61, $z = 1.75$ ($p = 0.08$), percentile block-bootstrap 95% CI $[-0.01, 1.21]$ (Seminar B13)
- Calmar has no usable sampling theory: its drawdown grows with the sample length ($\sqrt{T}$ for a driftless log price)

Caveats for Performance Measures

- All three ratios are estimates with large sampling error: $SE(\widehat{SR}) \approx 1/\sqrt{Y}$
- Heavy tails and negative skewness make the Sharpe ratio flattering
 - ▶ The Deflated Sharpe Ratio corrects for this and for multiple testing (Chapter 13)
- Different sample periods make comparisons across markets fragile
- Currency and interest rates: BET-TR is in RON, where short-term rates were well above USD rates; with $r_f = 0$ its Sharpe ratio is inflated
- Always report the ratios together with drawdowns and tail statistics

Risk and Return Across Markets



	CAGR	Sharpe	Sortino	Calmar
S&P 500	8.7%	0.55	0.78	0.15
BET-TR	21.2%	1.34	1.87	0.68
Bitcoin	53.9%	0.98	1.45	0.65
EUR/RON	1.8%	0.43	0.65	0.12
Gold	6.7%	0.48	0.69	0.15

- Different sample periods: not a horse race
- S&P 500 without dividends; BET-TR with dividends
- Sharpe ratios on simple returns; SEs (i.i.d. Normal) in the chart labels

 MFM_ch1_risk_measures

Stylised Facts in One Table

Fact	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Excess quantile kurtosis (Crow–Siddiqui > 0)	yes	yes	yes	yes	yes
Negative skewness	yes	strong	yes	no	yes
Volatility clustering (ARCH-LM)	yes	yes	yes	yes	yes
Slow decay of ACF $ r_t $	yes	weaker	weaker	yes	yes
Near-zero ACF of r_t	yes	small +	yes	no (+0.17)	yes
Leverage effect ($k = 1$)	-0.12	-0.12	-0.07	+0.10	-0.02

- Heavy tails, clustering and persistent volatility recur in all five samples; true long memory needs more evidence (breaks, Section 4)
- Autocorrelation and asymmetry depend on market microstructure and institutions

What a Good Model Must Reproduce

Requirements

- ▣ Time-varying, persistent volatility
- ▣ Heavy-tailed innovations, also after conditioning on volatility
- ▣ Asymmetric response to bad news
- ▣ Approximately uncorrelated returns

- ▣ The random walk with i.i.d. increments following the Normal distribution (Bachelier–Black–Scholes world) violates Facts 2, 3 and 5–11; Fact 4 holds trivially (sums of Normally distributed variables follow the Normal distribution)

Where in this course

- ▣ Ch. 5: GARCH; Ch. 6: multivariate volatility and dependence
- ▣ Ch. 7–8: VaR, Expected Shortfall, extreme value theory; backtesting
- ▣ Ch. 9: realised volatility from intraday data
- ▣ Ch. 13: machine learning must respect the low signal-to-noise ratio

Practical Checklist for Empirical Work

- ▣ **Know your data**
 - ▶ Price vs total-return index; adjusted vs raw prices; fixing vs traded price
 - ▶ Official reference rates where possible (e.g. EUR/RON)
- ▣ **Clean before you model**
 - ▶ Non-trading days (zero volume), holidays, stale quotes, obvious errors
 - ▶ Check extremes: are the largest moves real events?
- ▣ **Use log returns** for single-asset dynamics, simple returns for portfolios
- ▣ **Report robust statistics:** quantiles and tail counts next to kurtosis
- ▣ **Annualise carefully:** 252 vs 365 periods; the $\sqrt{P}$ rule fails under autocorrelation
- ▣ **Reproducibility:** code and data on GitHub, as Quantlets

Key Formulas

Returns

- ▣ $R_t = P_t/P_{t-1} - 1, \quad r_t = \ln(1 + R_t)$
- ▣ $r_t(k) = \sum_{j=0}^{k-1} r_{t-j}$
- ▣ $R_{p,t} = \sum_i w_i R_{i,t}$
- ▣ $\text{CAGR} = (P_T/P_0)^{1/Y} - 1$

Tests

- ▣ $JB = T[\hat{S}^2/6 + (\hat{K} - 3)^2/24]$
- ▣ Hill: $\text{SE}(\hat{\alpha}) \approx \hat{\alpha}/\sqrt{k}$
- ▣ $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k)$

Volatility

- ▣ $\sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$
- ▣ Parkinson: $\frac{1}{4 \ln 2} \overline{(h-l)^2}$
- ▣ Mixture: $K = 3 \mathbb{E}[\sigma^4] / \mathbb{E}[\sigma^2]^2$

Performance

- ▣ $SR = (\mu - r_f)/\sigma$ on simple returns; $\text{SE} \approx 1/\sqrt{Y}$
- ▣ $SoR = (\mu - r_f)/\sigma^-$
- ▣ $DD_t = P_t / \max_{s \leq t} P_s - 1$
- ▣ Calmar = $\text{CAGR} / |\text{MDD}|$

Case Study (1/7): Question

- Paper: [Guyon & Lekeufack \(2023\)](#), *Quantitative Finance*; the published results quoted below are from the authors' accepted manuscript (SSRN, revised 30 May 2023)
- Question: how much of an equity index's volatility can its past daily returns explain, and what function describes this relationship?
 - ▶ Facts 6, 8 and 9 of this chapter (clustering, slow decay, leverage) are usually measured one at a time; here a single formula has to produce all three
 - ▶ Builds on [Bollerslev, Hood, Huss & Pedersen \(2018\)](#): past realised variance averaged over several horizons forecasts volatility in every asset class; here the signed returns themselves are the only input
- Why it matters
 - ▶ If volatility is mostly a function of the price path, it is mostly endogenous: a path-dependent volatility (PDV) model needs little extra randomness, unlike stochastic volatility models (Chapters 11–12)
 - ▶ A volatility estimate from closing prices alone feeds VaR, volatility targeting and option pricing (Chapters 5, 7, 12)

Case Study (2/7): Data and Method

- Data: daily closes of the S&P 500, Euro Stoxx 50, FTSE 100, DAX and Nikkei 225 (Section 3)
 - ▶ Targets: the 30-day implied volatility index of each (VIX for the S&P 500) and the 9-day VIX; next-day realised volatility from 5-minute returns (Sections 3.4–3.5)
 - ▶ Training set 1 Jan 2000 – 31 Dec 2018; test set 1 Jan 2019 – 15 May 2022: calm 2019, the March 2020 crash and its aftermath
- Model, eq. (3.7), with simple daily returns $r_{t_i} = S_{t_i}/S_{t_{i-1}} - 1$

$$\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \Sigma_t, \quad R_{1,t} = \sum_{t_i \leq t} K_1(t - t_i) r_{t_i}, \quad \Sigma_t = \left(\sum_{t_i \leq t} K_2(t - t_i) r_{t_i}^2 \right)^{1/2}$$

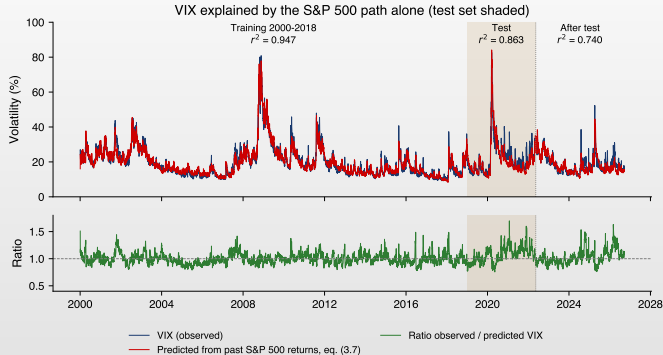
- ▶ σ_t : implied volatility at the close of day t , or realised volatility of day $t + 1$; $\beta_1 < 0$ gives the leverage effect, $\beta_2 \in (0, 1)$ gives clustering, as in GARCH
- ▶ Kernels: time-shifted power law (TSPL), eq. (3.9): $K(\tau) = Z_{\alpha, \delta}^{-1}(\tau + \delta)^{-\alpha}$, $\alpha > 1$, $\delta > 0$, normalised so that $\sum_i K(i\Delta t)\Delta t = 1$ with $\Delta t = 1/252$; 1,000 past days
- ▶ Large weight on the last few days (short memory) and a power-law tail (long memory, Fact 8)
- Estimation: the seven parameters $(\beta_0, \beta_1, \beta_2, \alpha_1, \delta_1, \alpha_2, \delta_2)$ by nonlinear least squares on the training set, initialised using a lasso regression on features constructed with exponential kernels; scores on each set (Section 3.4): RMSE $= ((\sigma_t - \hat{\sigma}_t)^2)^{1/2}$, $r^2 = 1 - \sum(\sigma_t - \hat{\sigma}_t)^2 / \sum(\sigma_t - \bar{\sigma})^2$, $\bar{\sigma}$ the mean of that set; volatility in decimals (0.03 = 3 volatility points)

Case Study (3/7): Results

Index	Implied volatility (Table 2)		Realised volatility (Table 4)	
	r^2 train	r^2 test	r^2 train	r^2 test
S&P 500 (VIX)	0.946	0.855	0.738	0.654
S&P 500 (9-day VIX)	0.876	0.914	—	—
Euro Stoxx 50	0.929	0.913	0.672	0.682
FTSE 100	0.925	0.870	0.650	0.617
DAX	0.934	0.918	0.722	0.557
Nikkei 225	0.890	0.800	0.563	0.504

- ▣ Past index returns alone explain 80–92% of the variance of implied volatility out of sample, with a test RMSE of about 3 volatility points (Table 2)
 - ▶ Interpretation: the VIX is mostly a function of the S&P 500 path; the exogenous part is small
- ▣ Realised volatility: test r^2 between 0.50 and 0.68 (Table 4), lower because 5-minute realised volatility is a noisy measure (Remark 3)
- ▣ VIX parameters: $\beta_0 = 0.057$, $\beta_1 = -0.095$, $\beta_2 = 0.82$; across all targets $\alpha_2 \in [1.2, 2.3]$ (Tables 3 and 5)
 - ▶ Interpretation: α_2 close to 1, the value below which volatility diverges, is long memory near criticality (Section 3.2)
- ▣ Dropping Σ_t hurts the most; the model beats two ARCH benchmarks and four published path-dependent models across indices and test sets (Figures 3.8–3.9, Appendix C)
- ▣ Its Markovian 4-factor version, driven by Brownian motion (local regularity $H = 1/2$), gives estimated Hurst exponents of about 0.12–0.16 for daily realised volatility, as in data (Section 4.6, Table 9): the roughness is only apparent at daily scales

Case Study (4/7): Replication

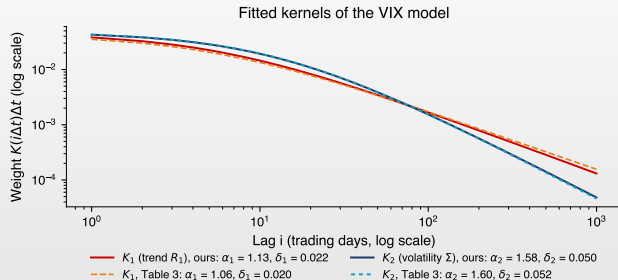


- Our fit of eq. (3.7) on GSPC.INDX and VIX.INDX: training 3 Jan 2000 – 31 Dec 2018 (4,779 days), started from Table 3, then parameters frozen
- $r^2 = 0.947$ on training and 0.863 on the test set, 2019 – 15 May 2022, shaded (Table 2: 0.946 and 0.855)

Case Study (5/7): Interpreting the Replication

- ▣ **Scores:** RMSE 0.020 on training and 0.034 on test, 2 Jan 2019 – 13 May 2022 (Table 2: 0.020 and 0.035)
 - ▶ The published result holds on the course data: the S&P 500 path alone explains 86% of the variance of the VIX out of sample
- ▣ **After the test set,** 16 May 2022 – 18 Sep 2026 (1,090 days), parameters frozen: $r^2 = 0.740$, RMSE 0.026
 - ▶ Interpretation: the RMSE is lower than on the test set, but the VIX varied less (standard deviation 5.0 versus 9.3 points), so the error is larger relative to its variability and $r^2 = 1 - \text{RMSE}^2 / \text{Var}$ falls; whether the relationship itself changed needs a separate analysis
- ▣ **Ratio of observed to predicted VIX** (lower panel): mean-reverts around 1, the exogenous part (Section 5)
 - ▶ Mean 0.85 in April–May 2020 and 1.16 from October 2020 to May 2022, the two episodes of Figure 3.2
 - ▶ 8 April 2025, the tariff shock: VIX 52.3 versus 39.2 predicted, a ratio of 1.34

Case Study (6/7): Fitted Kernels



- Daily weights $K(i\Delta t)\Delta t$ of our fit (solid) and of Table 3 (dashed), log-log scale; slope $\approx -\alpha$ for lags well beyond $\delta/\Delta t$ days
- K_2 : $\hat{\alpha}_2 = 1.58$, $\hat{\delta}_2 = 0.050$ versus 1.60, 0.052 in Table 3: the same curve
 - The last 5 days carry 20% of the weight and the days beyond lag 250 still 10%: short and long memory in one kernel (Facts 6 and 8)
- K_1 : $\hat{\alpha}_1 = 1.13$ versus 1.06 and $\hat{\beta}_1 = -0.090$ versus -0.095 ; $\hat{\beta}_2 = 0.868$ versus 0.82
 - Both kernels are normalised on the 1,000-day window, as for Table 3, so the β are on the same scale; parameter differences alone do not measure precision (that needs standard errors or a bootstrap)

Case Study (7/7): Replication and Extension

▣ Replication with course data (data/market)

- ▶ GSPC.INDX and VIX.INDX cover the published training and test periods; VIX9D.INDX starts in 2011 (shorter training set)
- ▶ VIX fit of eq. (3.7) from the last 1,000 returns: slides 4/7–6/7; the 9-day VIX and the restricted models ($\beta_1 = 0$, $\beta_2 = 0$) are left to Seminar 1, C3
- ▶ No 5-minute data in the course: Table 4 cannot be replicated exactly; the range estimators of this chapter are a noisier target

▣ Limits stated by the authors

- ▶ Realised volatility targets are noisy, so their scores understate the endogenous share (Remark 3)
- ▶ A small exogenous part remains: the ratio of observed to fitted VIX mean-reverts around 1 and needs a stochastic factor (Section 5)
- ▶ On the test set the VIX fell faster than predicted in April–May 2020 and stayed higher than predicted after October 2020 (Figure 3.2)

- ▣ **Open extension:** the frozen VIX fit keeps a low RMSE after May 2022 (slides 4/7 and 5/7), yet misses part of the April 2025 tariff shock; does the same formula describe realised volatility of Bitcoin or of the BET index?

An Open Question: Are Tail Indices Stable over Time?

- The stylised facts are usually stated for a whole sample; the tail index α is treated as a constant
- Crypto-assets changed in structure: futures (2017), spot Bitcoin ETFs (January 2024), institutional holders
 - ▶ [Pele et al. \(2023\)](#): the tail factor is what separates cryptos from other asset classes
- Question: did institutionalisation make Bitcoin's tails thinner, i.e. raise α ?
- Inference is delicate: $\hat{\alpha}$ uses only the k largest observations, and returns are dependent
 - ▶ [Hoga \(2017\)](#): change-point tests for the tail index valid under β -mixing (e.g. GARCH-type) dependence

The Discovery Loop, Step by Step (1/2)

□ Literature

- ▶ Find peer-reviewed work since 2015 on time variation or structural breaks in tail indices of asset returns, with tests valid under GARCH-type dependence. For each: DOI, estimator, test, data. Flag entries you are unsure of.
- ▶ Resolve every DOI on Crossref; the title and authors must match

□ Hypothesis

- ▶ State competing hypotheses for a change in Bitcoin's tail after the 2024 spot ETFs: a change in tail shape, a fall in volatility only, sampling noise. What does each predict for the Hill index of raw returns and of GARCH-standardised returns?

□ Data and code

- ▶ Read BTC-USD.CC from data/market (close); log returns on its own calendar; Hill index of $|r|$ with $k = 2.5\%$ of n before and after 11 January 2024; moving-block bootstrap standard errors.
- ▶ Replicate a known number first: the chapter's Hill estimates, Bitcoin 4.2 and S&P 500 3.1

The Discovery Loop, Step by Step (2/2)

▣ Robustness

- ▶ Write a pre-registered robustness plan: k from 1% to 5% of n , loss and gain tails separately, GARCH-standardised residuals, block lengths from 10 to 50 days, a placebo date and a placebo asset.

▣ Critique

- ▶ Referee this design. With 24 tail observations after the break, what is the power against a change of 1 in the tail index? Which dependence assumption does each standard error need?

▣ Report

- ▶ $\hat{\alpha}$ in each window, both standard errors, the placebo, the minimum detectable change
- ▶ Prompts that shaped a decision go into AI_USE.md

- ▣ Scale versus shape: a lower volatility leaves α unchanged; only a change in the shape of the tail moves it

What the Researcher Must Check

- **Wrong return or price series:** prices, simple returns or a different calendar
 - ▶ test: reproduce the chapter's Hill table before anything else
- **Choice of k after seeing the result:** fix the rule ($k = 2.5\%$ of n) in advance; show the Hill plot for 1–5%
- **i.i.d. standard errors:** $\hat{\alpha}/\sqrt{k}$ ignores volatility clustering
 - ▶ test: compare with a block bootstrap; for the S&P 500 before 2024 it is 2.2 times larger
- **Break date chosen from the data:** use a date fixed by the event, or a sup-type change-point test with its own critical values ([Hoga \(2017\)](#))
- **Bad ticks create fake tails:** list the ten largest $|r_t|$ and match each to a dated event
- **Low power:** report the minimum detectable change: 2.8 standard errors of the difference for a two-sided 5% test with 80% power, 2.5 for a one-sided test

Mini-Case: Did Bitcoin's Tail Get Thinner after the ETFs?

- A typical assistant proposal: “institutional money since the spot ETFs has tamed Bitcoin's tails”
- Test: Hill index of $|r_t|$, $k = 2.5\%$ of n ; $H_0 : \alpha_{\text{post}} = \alpha_{\text{pre}}$ against $H_1 : \alpha_{\text{post}} > \alpha_{\text{pre}}$
 - ▶ $z = (\hat{\alpha}_{\text{post}} - \hat{\alpha}_{\text{pre}}) / \sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2}$, s from a moving-block bootstrap (blocks of 20 days)
- Before (Sep 2014 – 10 Jan 2024, $k = 85$): $\hat{\alpha} = 4.16$ ($s = 0.52$)
- After (11 Jan 2024 – Sep 2026, $k = 24$): $\hat{\alpha} = 4.31$ ($s = 0.91$); $z = 0.15$
- Placebo, S&P 500 at the same date: $2.82 \rightarrow 2.86$, $z = 0.03$
- Interpretation: no evidence of a change, but for this one-sided 5% test with 80% power the minimum detectable change is $(1.645 + 0.842) \times 1.05 = 2.6$
 - ▶ a normal approximation, rough with 24 tail observations: the data cannot distinguish “unchanged” from a large thinning of the tail; wait for more data or pool assets

 ai_discovery_case.py

Your Turn: A Project Seed

- **Question:** did the tail index of Bitcoin and Ethereum change with institutionalisation (futures 2017, spot ETFs 2024)?
 - ▶ raw and GARCH-standardised returns: shape versus scale; loss and gain tails separately
- **Replicate first:** the chapter's Hill table ($k = 2.5\%$ of n) and the mini-case numbers
- **Deliverables**
 - ▶ a pre-registration note (H_0 , H_1 , rule for k , break dates, bootstrap) committed before the results
 - ▶ the change-point test of [Hoga \(2017\)](#) as a Quantlet; a report with power calculations
- **Logs:** prompts in `AI_USE.md`; every caught error (an i.i.d. standard error, a date chosen from the data) in `AI_ERRORS.md`

Conclusions

- ▣ Returns are **uncorrelated but not independent**: their size is predictable, their linear autocorrelation is weak
- ▣ Heavy tails are the rule: 5σ days are 1000+ times more frequent than the Normal distribution predicts
- ▣ Volatility clusters, is highly persistent (long memory or breaks) and reacts asymmetrically to losses in equity markets
- ▣ Tails thin out with aggregation, but slowly
- ▣ Crypto shows the same facts with larger amplitude; managed exchange rates and thin markets add autocorrelation
- ▣ Range-based estimators extract more information from OHLC data
- ▣ Risk-adjusted measures must be read together with drawdowns and tail statistics

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Statistics of Financial Markets
- ▣  Tukey's g- and h- transformations
- ▣  Applied Time Series Analysis with Python

Code and data

- ▣ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_01

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